

Complex Numbers

ANSWER KEY



Powers of i

- 1 Evaluate each power of i , using $i^2 = -1$ and the fact that powers of i repeat every 4:

a $i^3 = -i$

b $i^4 = 1$

c $i^7 = -i$

d $i^{22} = -1$

- 2 Evaluate i^{101} .

$$101 = 4(25) + 1, \text{ so } i^{101} = (i^4)^{25} \cdot i^1 = 1 \cdot i = i.$$

Addition and subtraction

- 3 Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

$$= 1 + 9i$$

- 4 Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

$$= 1 - 7i$$

- 5 Find $(7 + 2i) - (3 - 5i)$.

$$= 4 + 7i$$

- 6 Find $(4 - i) - (-2 + 6i)$.

$$= 6 - 7i$$

Multiplication

- 7 Multiply $(2 + 3i)(1 - 4i)$.

$$(2)(1) - (3)(-4) + [(2)(-4) + (3)(1)]i = 14 - 5i$$

- 8 Multiply $(-1 + 2i)(3 + i)$.

$$(-1)(3) - (2)(1) + [(-1)(1) + (2)(3)]i = -5 + 5i$$

- 9 Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.

$$= 25 - (-4) + 0i = 29. \text{ It is a product of conjugates } z\bar{z} = |z|^2 \text{ (Theorem 1.1), which is always real.}$$

Conjugate and modulus

- 10 For $z = 6 - 8i$, find $\bar{z}$ and $|z|$.

$$\bar{z} = 6 + 8i, \text{ and } |z| = \sqrt{6^2 + 8^2} = \sqrt{100} = 10.$$

- 11 For $z = -5 + 12i$, find $\bar{z}$ and $|z|$.

$$\bar{z} = -5 - 12i, \text{ and } |z| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

- 12 Show that $z + \bar{z}$ is always real for $z = a + bi$.

$$z + \bar{z} = (a + bi) + (a - bi) = 2a, \text{ which is real since } a \text{ is real.}$$

Division

- 13 Simplify $\frac{3 + 4i}{1 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

$$\text{Multiply by } \frac{1 + 2i}{1 + 2i}: \text{ numerator} = (3)(1) - (4)(2) + [(3)(2) + (4)(1)]i = -5 + 10i; \text{ denominator} = 1^2 + 2^2 = 5. \text{ So the quotient is } -1 + 2i.$$

- 14 Simplify $\frac{1 + i}{1 - i}$.

$$\text{Multiply by } \frac{1 + i}{1 + i}: \text{ numerator} = (1)(1) - (1)(1) + [(1)(1) + (1)(1)]i = 0 + 2i; \text{ denominator} = 1^2 + 1^2 = 2. \text{ So the quotient is } i.$$

- 15 Simplify $\frac{7 - i}{3 + 2i}$.

$$\text{Multiply by } \frac{3 - 2i}{3 - 2i}: \text{ numerator} = (7)(3) - (-1)(-2) + [(7)(-2) + (-1)(3)]i = 19 - 17i; \text{ denominator} = 3^2 + 2^2 = 13. \text{ So the quotient is } \frac{19}{13} - \frac{17}{13}i.$$

Quadratic equations with complex roots

- 16 Solve $x^2 + 9 = 0$.

$$x^2 = -9 \Rightarrow x = \pm 3i.$$

- 17 Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

$$\text{Discriminant} = 36 - 52 = -16, \text{ so } \sqrt{-16} = 4i. x = \frac{6 \pm 4i}{2} = 3 \pm 2i.$$

- 18 Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.

$$\text{Discriminant} = 4 - 40 = -36, \text{ so } \sqrt{-36} = 6i. x = \frac{-2 \pm 6i}{4} = -\frac{1}{2} \pm \frac{3}{2}i.$$

Multiple choice — easy

19 Evaluate i^{130} , using the fact that powers of i repeat every 4.

- a. i
- b. $-i$
- c. 1
- d. -1 ✓

$$130 = 4(32) + 2, \text{ so } i^{130} = (i^4)^{32} \cdot i^2 = 1 \cdot -1 = -1.$$

20 Evaluate i^{160} , using the fact that powers of i repeat every 4.

- a. -1
- b. $-i$
- c. 1 ✓
- d. i

$$160 = 4(40), \text{ so } i^{160} = (i^4)^{40} \cdot i^0 = 1 \cdot 1 = 1.$$

21 Evaluate i^{211} , using the fact that powers of i repeat every 4.

- a. 1
- b. -1
- c. $-i$ ✓
- d. i

$$211 = 4(52) + 3, \text{ so } i^{211} = (i^4)^{52} \cdot i^3 = 1 \cdot -i = -i.$$

22 Evaluate i^{236} , using the fact that powers of i repeat every 4.

a. 1 ✓

b. -1

c. i

d. $-i$

$$236 = 4(59), \text{ so } i^{236} = (i^4)^{59} \cdot i^0 = 1 \cdot 1 = 1.$$

23 Evaluate i^{229} , using the fact that powers of i repeat every 4.

a. 1

b. i ✓

c. -1

d. $-i$

$$229 = 4(57) + 1, \text{ so } i^{229} = (i^4)^{57} \cdot i^1 = 1 \cdot i = i.$$

24 Evaluate i^{52} , using the fact that powers of i repeat every 4.

a. -1

b. 1 ✓

c. $-i$

d. i

$$52 = 4(13), \text{ so } i^{52} = (i^4)^{13} \cdot i^0 = 1 \cdot 1 = 1.$$

25 Evaluate i^{206} , using the fact that powers of i repeat every 4.

a. i

b. $-i$

c. -1 ✓

d. 1

$$206 = 4(51) + 2, \text{ so } i^{206} = (i^4)^{51} \cdot i^2 = 1 \cdot -1 = -1.$$

26 Evaluate i^{81} , using the fact that powers of i repeat every 4.

a. -1

b. $-i$

c. 1

d. i ✓

$$81 = 4(20) + 1, \text{ so } i^{81} = (i^4)^{20} \cdot i^1 = 1 \cdot i = i.$$

27 Evaluate i^{62} , using the fact that powers of i repeat every 4.

a. i

b. -1 ✓

c. 1

d. $-i$

$$62 = 4(15) + 2, \text{ so } i^{62} = (i^4)^{15} \cdot i^2 = 1 \cdot -1 = -1.$$

28 Evaluate i^{231} , using the fact that powers of i repeat every 4.

a. 1

b. -1

c. $-i$ ✓

d. i

$$231 = 4(57) + 3, \text{ so } i^{231} = (i^4)^{57} \cdot i^3 = 1 \cdot -i = -i.$$

29 Evaluate i^{101} , using the fact that powers of i repeat every 4.

a. $-i$

b. -1

c. i ✓

d. 1

$$101 = 4(25) + 1, \text{ so } i^{101} = (i^4)^{25} \cdot i^1 = 1 \cdot i = i.$$

30 Evaluate i^{66} , using the fact that powers of i repeat every 4.

a. i

b. 1

c. $-i$

d. -1 ✓

$$66 = 4(16) + 2, \text{ so } i^{66} = (i^4)^{16} \cdot i^2 = 1 \cdot -1 = -1.$$

31 Evaluate i^{31} , using the fact that powers of i repeat every 4.

a. -1

b. i

c. 1

d. $-i$ ✓

$$31 = 4(7) + 3, \text{ so } i^{31} = (i^4)^7 \cdot i^3 = 1 \cdot -i = -i.$$

32 Evaluate i^7 , using the fact that powers of i repeat every 4.

a. $-i$ ✓

b. 1

c. i

d. -1

$$7 = 4(1) + 3, \text{ so } i^7 = (i^4)^1 \cdot i^3 = 1 \cdot -i = -i.$$

33 Evaluate i^{178} , using the fact that powers of i repeat every 4.

a. i

b. 1

c. -1 ✓

d. $-i$

$$178 = 4(44) + 2, \text{ so } i^{178} = (i^4)^{44} \cdot i^2 = 1 \cdot -1 = -1.$$

Multiple choice — medium

34 Multiply $(-9 - 4i)(1 - 6i)$.

a. $-33 + 15i$

b. $-33 + 50i$ ✓

c. $15 + 50i$

d. $-33 + 58i$

$$(-9)(1) - (-4)(-6) + [(-9)(-6) + (-4)(1)]i = -33 + 50i.$$

35 Multiply $(3 - i)(-3 - 9i)$.

a. $-18 - 24i$ ✓

b. $-18 - 30i$

c. $-24i$

d. -18

$$(3)(-3) - (-1)(-9) + [(3)(-9) + (-1)(-3)]i = -18 - 24i.$$

36 Multiply $(-3 - 2i)(-3 - 2i)$.

a. $5 + 13i$

b. $5 + 12i$ ✓

c. $13 + 12i$

d. 5

$$(-3)(-3) - (-2)(-2) + [(-3)(-2) + (-2)(-3)]i = 5 + 12i.$$

37 Multiply $(7 + 2i)(4)$.

a. $29 + 8i$

b. $28 + 8i$ ✓

c. $28 - 8i$

d. $28 + 28i$

$$(7)(4) - (2)(0) + [(7)(0) + (2)(4)]i = 28 + 8i.$$

38 Multiply $(-8 - 9i)(-5 - i)$.

a. $49 + 53i$

b. $31 + 49i$

c. $31 + 53i$ ✓

d. $31 - 37i$

$$(-8)(-5) - (-9)(-1) + [(-8)(-1) + (-9)(-5)]i = 31 + 53i.$$

39 Multiply $(7 - 6i)(4 + 8i)$.

a. $-20 + 32i$

b. $76 + 80i$

c. $76 - 20i$

d. $76 + 32i$ ✓

$$(7)(4) - (-6)(8) + [(7)(8) + (-6)(4)]i = 76 + 32i.$$

40 Multiply $(3 - 9i)(-8 + 3i)$.

a. $3 - 51i$

b. $3 - 63i$

c. $3 + 81i$ ✓

d. $-51 + 81i$

$$(3)(-8) - (-9)(3) + [(3)(3) + (-9)(-8)]i = 3 + 81i.$$

41 Multiply $(-3 - 7i)(1 - 3i)$.

a. $-24 + 18i$

b. $-24 + 2i$ ✓

c. $18 + 2i$

d. $-24 + 16i$

$$(-3)(1) - (-7)(-3) + [(-3)(-3) + (-7)(1)]i = -24 + 2i.$$

42 Multiply $(3 + 6i)(5 + 6i)$.

a. $51 + 48i$

b. $-21 + 48i$ ✓

c. $-21 - 12i$

d. $-21 + 51i$

$$(3)(5) - (6)(6) + [(3)(6) + (6)(5)]i = -21 + 48i.$$

43 Multiply $(8 - 5i)(4 + 6i)$.

- a. $62 + 2i$
- b. $62 + 68i$
- c. $62 + 28i$ ✓
- d. $2 + 28i$

$$(8)(4) - (-5)(6) + [(8)(6) + (-5)(4)]i = 62 + 28i.$$

44 Multiply $(8 - 5i)(9 + 4i)$.

- a. $92 + 52i$
- b. $52 - 13i$
- c. $92 - 13i$ ✓
- d. $92 + 77i$

$$(8)(9) - (-5)(4) + [(8)(4) + (-5)(9)]i = 92 - 13i.$$

45 Multiply $(-i)(-3 + 8i)$.

- a. $8 - 3i$
- b. $8 - 8i$
- c. $-8 + 3i$
- d. $8 + 3i$ ✓

$$(0)(-3) - (-1)(8) + [(0)(8) + (-1)(-3)]i = 8 + 3i.$$

46 Multiply $(2 - 2i)(-6 - 5i)$.

- a. $-22 - 2i$
- b. $-22 + 2i$ ✓
- c. $-2 + 2i$
- d. $-22 - 22i$

$$(2)(-6) - (-2)(-5) + [(2)(-5) + (-2)(-6)]i = -22 + 2i.$$

47 Multiply $(-1 - 2i)(9 - 4i)$.

- a. $-17 + 22i$
- b. $-17 - i$
- c. $-1 - 14i$
- d. $-17 - 14i$ ✓

$$(-1)(9) - (-2)(-4) + [(-1)(-4) + (-2)(9)]i = -17 - 14i.$$

48 Multiply $(-7 - 6i)(2 + 3i)$.

- a. $4 - 9i$
- b. $4 - 33i$ ✓
- c. $4 - 32i$
- d. $-32 - 33i$

$$(-7)(2) - (-6)(3) + [(-7)(3) + (-6)(2)]i = 4 - 33i.$$

Multiple choice — hard

49 Simplify $\frac{-34 - 8i}{-7 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

- a. $-12/53 + 254/53i$
- b. $307/53 - 12/53i$
- c. $254/53 - 12/53i$ ✓
- d. $254/53 + 12/53i$

Multiply by $(-7 + 2i)/(-7 + 2i)$: numerator

$$= (-34)(-7) - (-8)(2) + [(-8)(-7) + (-34)(2)]i = 254 - 12i; \text{ denominator } = (-7)^2 + (-2)^2 = 53.$$

Quotient: $254/53 - 12/53i$.

- 50 Simplify $\frac{6+8i}{2+8i}$ by multiplying top and bottom by the conjugate of the denominator.

- a. $-8/17 + 19/17i$
- b. $19/17 + 8/17i$
- c. $19/17 - 8/17i$ ✓
- d. $36/17 - 8/17i$

Multiply by $(2-8i)/(2-8i)$: numerator = $(6)(2) - (8)(-8) + [(8)(2) + (6)(-8)]i = 76 - 32i$;
denominator = $(2)^2 + (8)^2 = 68$. Quotient: $19/17 - 8/17i$.

- 51 Simplify $\frac{-10-54i}{-1-9i}$ by multiplying top and bottom by the conjugate of the denominator.

- a. $248/41 - 18/41i$ ✓
- b. $289/41 - 18/41i$
- c. $248/41 + 18/41i$
- d. $-18/41 + 248/41i$

Multiply by $(-1+9i)/(-1+9i)$: numerator
= $(-10)(-1) - (-54)(9) + [(-54)(-1) + (-10)(9)]i = 496 - 36i$; denominator
= $(-1)^2 + (-9)^2 = 82$. Quotient: $248/41 - 18/41i$.

- 52 Simplify $\frac{16+4i}{4+2i}$ by multiplying top and bottom by the conjugate of the denominator.

- a. $18/5 - 4/5i$ ✓
- b. $-4/5 + 18/5i$
- c. $18/5 + 4/5i$
- d. $23/5 - 4/5i$

Multiply by $(4-2i)/(4-2i)$: numerator = $(16)(4) - (4)(-2) + [(4)(4) + (16)(-2)]i = 72 - 16i$;
denominator = $(4)^2 + (2)^2 = 20$. Quotient: $18/5 - 4/5i$.

53 Simplify $\frac{-45 - 12i}{-8 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $3/34 + 96/17i$

b. $96/17 + 3/34i$ ✓

c. $96/17 - 3/34i$

d. $113/17 + 3/34i$

Multiply by $(-8 + 2i)/(-8 + 2i)$: numerator

$$= (-45)(-8) - (-12)(2) + [(-12)(-8) + (-45)(2)]i = 384 + 6i; \text{ denominator}$$

$$= (-8)^2 + (-2)^2 = 68. \text{ Quotient: } 96/17 + 3/34i.$$

54 For $z = 9 - 40i$, find $|z|$.

a. 49

b. 42

c. 40

d. 41 ✓

$$|z| = \sqrt{(9)^2 + (-40)^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41.$$

55 For $z = 3 - 4i$, find $|z|$.

a. 5 ✓

b. 4

c. 6

d. 7

$$|z| = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

56 For $z = 5 + 12i$, find $|z|$.

- a. 17
- b. 12
- c. 14
- d. 13 ✓

$$|z| = \sqrt{(5)^2 + (12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

57 For $z = 7 + 24i$, find $|z|$.

- a. 26
- b. 25 ✓
- c. 24
- d. 31

$$|z| = \sqrt{(7)^2 + (24)^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

58 For $z = 8 - 15i$, find $|z|$.

- a. 17 ✓
- b. 18
- c. 16
- d. 23

$$|z| = \sqrt{(8)^2 + (-15)^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

59 Solve $x^2 + 14x + 98 = 0$ using the quadratic formula.

a. $-7 + 7i, -7 - 7i$ ✓

b. $-6 + 7i, -6 - 7i$

c. $-7 + 7i$

d. $7 + 7i, 7 - 7i$

Discriminant $= (14)^2 - 4(98) = 196 - 392 = -196$, so $\sqrt{-196} = 14i$.
 $x = (-14 \pm 14i)/2 = -7 + 7i, -7 - 7i$.

60 Solve $x^2 - 14x + 113 = 0$ using the quadratic formula.

a. $7 + 8i$

b. $-7 + 8i, -7 - 8i$

c. $8 + 8i, 8 - 8i$

d. $7 + 8i, 7 - 8i$ ✓

Discriminant $= (-14)^2 - 4(113) = 196 - 452 = -256$, so $\sqrt{-256} = 16i$.
 $x = (14 \pm 16i)/2 = 7 + 8i, 7 - 8i$.

61 Solve $x^2 + 25 = 0$ using the quadratic formula.

a. $5i, -5i$ ✓

b. $5i$

c. $1 + 5i, 1 - 5i$

d. $2 + 5i, 2 - 5i$

Discriminant $= (0)^2 - 4(25) = 0 - 100 = -100$, so $\sqrt{-100} = 10i$. $x = (0 \pm 10i)/2 = 5i, -5i$.

62 Solve $x^2 + 14x + 85 = 0$ using the quadratic formula.

a. $-7 + 6i$

b. $-7 + 6i, -7 - 6i$ ✓

c. $-6 + 6i, -6 - 6i$

d. $7 + 6i, 7 - 6i$

Discriminant $= (14)^2 - 4(85) = 196 - 340 = -144$, so $\sqrt{-144} = 12i$.
 $x = (-14 \pm 12i)/2 = -7 + 6i, -7 - 6i$.

63 Solve $x^2 + 16 = 0$ using the quadratic formula.

a. $1 + 4i, 1 - 4i$

b. $4i$

c. $2 + 4i, 2 - 4i$

d. $4i, -4i$ ✓

Discriminant $= (0)^2 - 4(16) = 0 - 64 = -64$, so $\sqrt{-64} = 8i$. $x = (0 \pm 8i)/2 = 4i, -4i$.